

Knowledge Distillation — Training Report

Generated from JSON logs in `results/training_logs`. This report inspects *training dynamics and checkpoint quality* — final test metrics, parameter counts, GFLOPs and latency live in the separate evaluation report.

Checkpoint selection: teachers, CE baselines and MSE+CE students are selected by highest validation Top-1; MSE-only students by lowest validation MSE (Top-1 can be a zero placeholder during training and is not meaningful for them).

Experiment Configuration

Training hyperparameters loaded directly from `configs/*.yaml`.

Parameter	aircraft_convnext_tiny	aircraft_resnet50	food101_convnext_tiny	food101_resnet50
image_size	224	224	224	224
batch_size	32	32	32	32
num_workers	4	4	4	4
stage1_epochs	30	30	30	30
stage2_epochs	15	15	15	15
stage2_lr	5e-05	5e-05	5e-05	5e-05
epochs	30	30	30	30
lr	0.001	0.001	0.001	0.001
weight_decay	0.0001	0.0001	0.0001	0.0001
scheduler	cosine	cosine	cosine	cosine
alpha	1.0	1.0	1.0	1.0
beta	1.0	1.0	1.0	1.0
amp	True	True	True	True
seed	42	42	42	42

Executive Summary

Dataset	Best teacher	T.T1	Best baseline	B.T1	Best distilled (T1)	D.T1	Lowest MSE run	MSE	High-concern
aircraft	CvN	58.84	S-L base	35.61	CvN/S-L/pre/MSE+CE	55.15	RN50/S-L/post/MSE	0.0858	0
food101	CvN	78.36	S-L base	64.66	CvN/S-L/pre/MSE+CE	62.93	CvN/S-L/post/MSE	0.0331	1

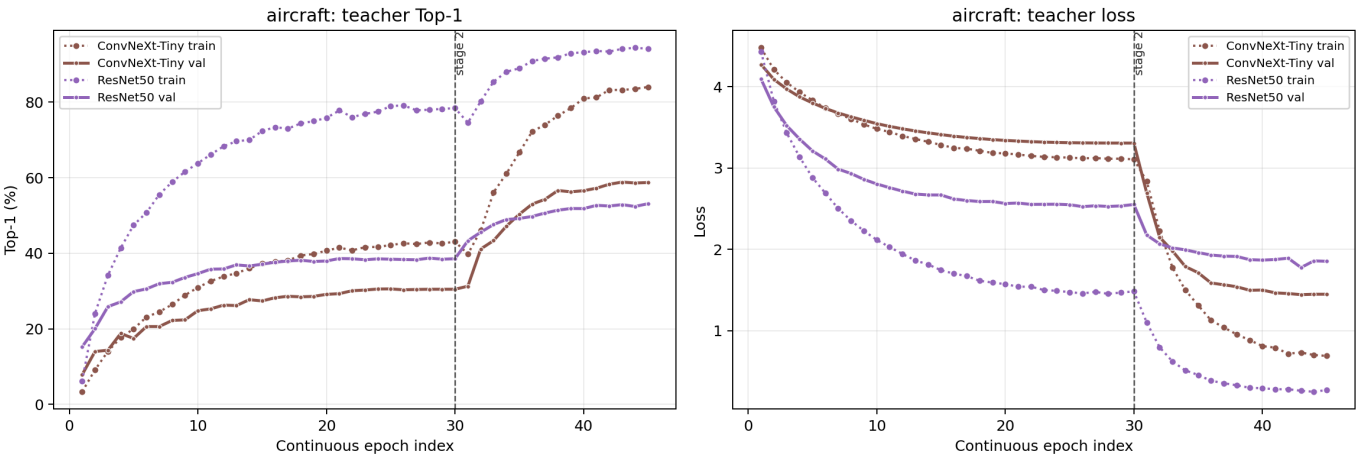
Teacher Training

Teacher adaptation is two-stage: the dataset head is trained with the encoder frozen, then the last encoder block is fine-tuned. The dashed line in the plots marks that transition. **Best@ is the continuous epoch index**, so a stage-2 best shows its true position (e.g. 45), not the local stage-2 counter (e.g. 15).

aircraft

Teacher	Ep	Best@	Stg	Best T1	Fin T1	T1 drop	Best L	Fin L	Concern
convnext_tiny	45	43	2	58.84	58.72	0.12	1.44	1.45	medium
resnet50	45	45	2	53.11	53.11	0.00	1.85	1.85	medium

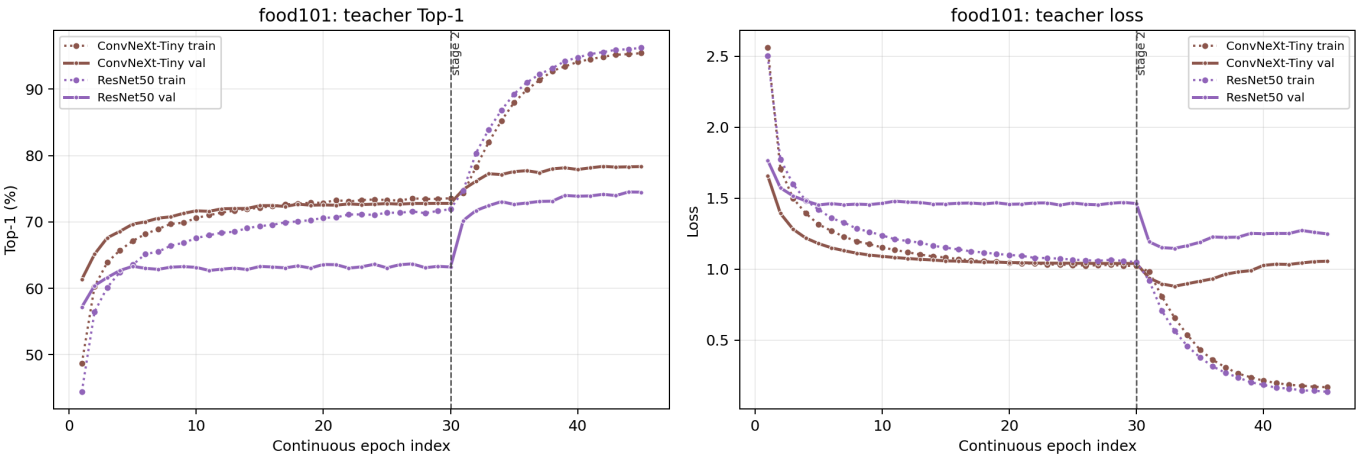
Dashed line = transition from frozen-head training to last-block fine-tuning



food101

Teacher	Ep	Best@	Stg	Best T1	Fin T1	T1 drop	Best L	Fin L	Concern
convnext_tiny	45	42	2	78.36	78.34	0.03	1.03	1.06	high
resnet50	45	44	2	74.51	74.48	0.03	1.26	1.25	medium

Dashed line = transition from frozen-head training to last-block fine-tuning

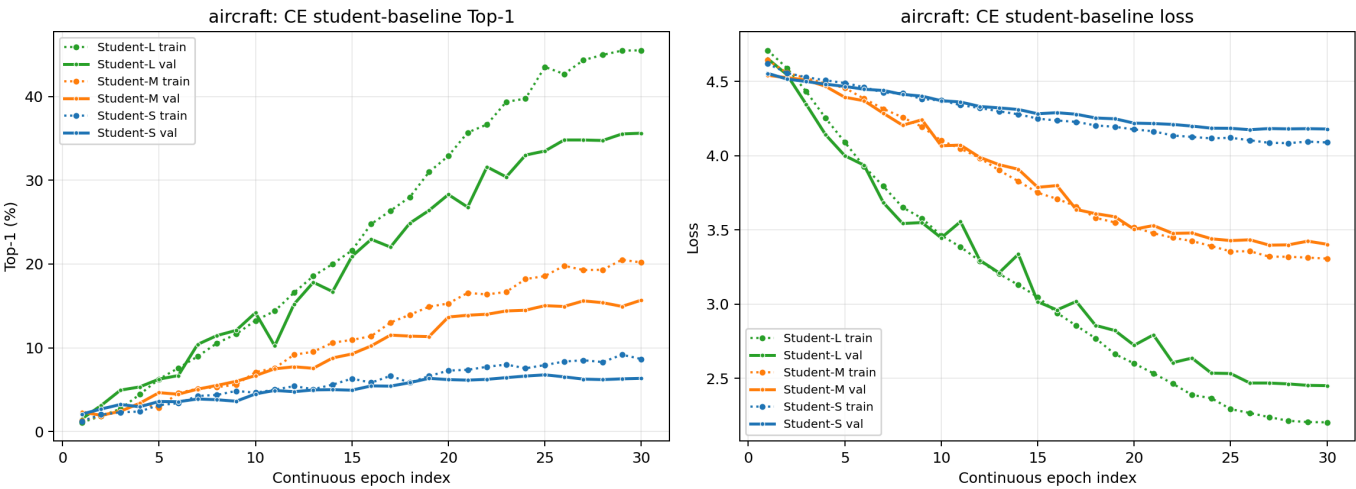


Student Baselines (CE)

Supervised CE students are the reference distillation must beat. **vs base** columns elsewhere are measured against these.

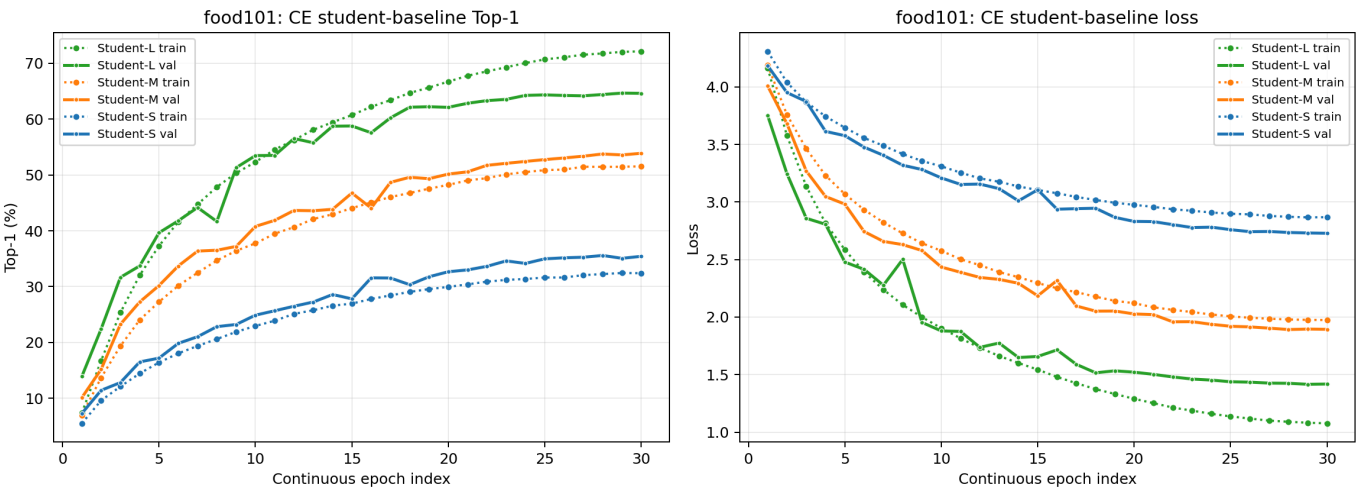
aircraft

Student	Ep	Best@	Best T1	Fin T1	T1 drop	Best L	Fin L	Concern
student_l	30	30	35.61	35.61	0.00	2.45	2.45	low
student_m	30	30	15.69	15.69	0.00	3.40	3.40	low
student_s	30	25	6.78	6.36	0.42	4.18	4.18	low



food101

Student	Ep	Best@	Best T1	Fin T1	T1 drop	Best L	Fin L	Concern
student_l	30	29	64.66	64.62	0.04	1.42	1.42	low
student_m	30	30	53.87	53.87	0.00	1.89	1.89	low
student_s	30	28	35.58	35.42	0.16	2.74	2.73	low



Distillation — aircraft

- Best classification run: [convnext_tiny/student_l/pregap/mse_ce](#) — 55.146% val Top-1.
- Lowest-MSE run: [resnet50/student_l/postgap/mse](#) — 0.0858 val MSE.

All distilled runs

Teacher	Student	Target	Loss	Ep	Best@	Best T1	Fin T1	T1 drop	Best MSE	Fin MSE	MSE reg%	T1 gap	vs base	Concern
convnext_tiny	student_l	postgap	mse	30	28	0.00	0.00	0.00	0.2220	0.2221	0.1	0.00		low
convnext_tiny	student_l	postgap	mse_ce	30	28	26.04	24.96	1.08	0.3879	0.4247	9.6	2.75	-9.57	low
convnext_tiny	student_l	pregap	mse	30	30	0.00	0.00	0.00	1.0053	1.0053	0.0	0.00		low
convnext_tiny	student_l	pregap	mse_ce	30	29	55.15	54.85	0.30	1.4622	1.4642	0.1	24.01	19.53	medium
convnext_tiny	student_m	postgap	mse	30	30	0.00	0.00	0.00	0.2439	0.2439	0.0	0.00		low
convnext_tiny	student_m	postgap	mse_ce	30	29	24.75	24.42	0.33	0.3889	0.3951	2.4	2.69	9.06	low
convnext_tiny	student_m	pregap	mse	30	30	0.00	0.00	0.00	1.2041	1.2041	0.0	0.00		low
convnext_tiny	student_m	pregap	mse_ce	30	29	40.65	40.17	0.48	1.5860	1.5937	0.6	12.59	24.96	low

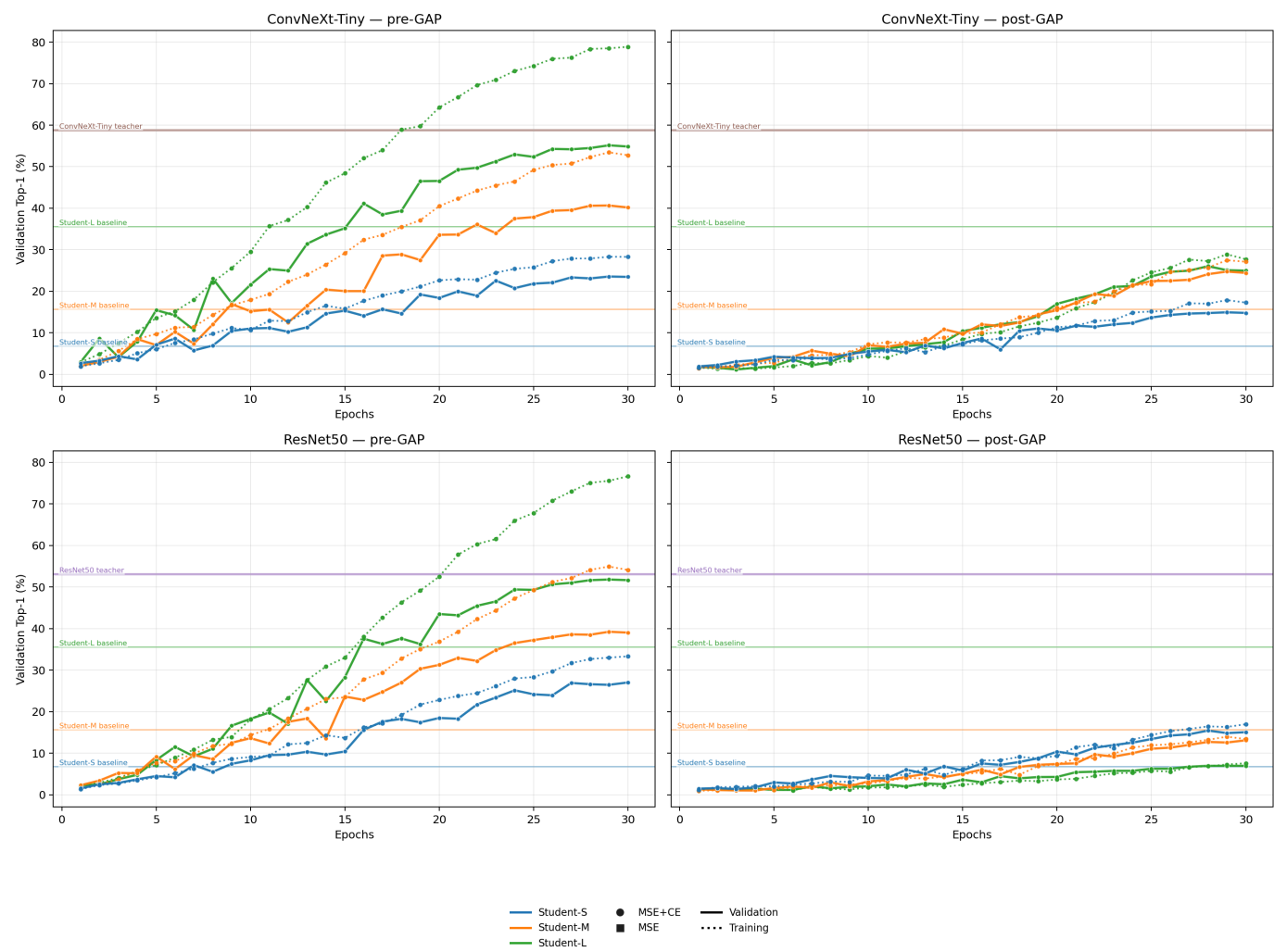
Teacher	Student	Target	Loss	Ep	Best@	Best T1	Fin T1	T1 drop	Best MSE	Fin MSE	MSE reg%	T1 gap	vs base	Concern
convnext_tiny	student_s	postgap	mse	30	30	0.00	0.00	0.00	0.2634	0.2634	0.0	0.00		low
convnext_tiny	student_s	postgap	mse_ce	30	29	14.94	14.79	0.15	0.4453	0.4530	4.2	2.49	8.16	low
convnext_tiny	student_s	pregap	mse	30	27	0.00	0.00	0.00	1.3816	1.3821	0.0	0.00		low
convnext_tiny	student_s	pregap	mse_ce	30	29	23.55	23.46	0.09	1.6637	1.6655	0.2	4.82	16.77	low
resnet50	student_l	postgap	mse	30	29	0.00	0.00	0.00	0.0858	0.0858	0.0	0.00		low
resnet50	student_l	postgap	mse_ce	30	28	6.96	6.96	0.00	0.1635	0.1665	2.3	0.63	-28.65	low
resnet50	student_l	pregap	mse	30	29	0.00	0.00	0.00	1.0475	1.0475	0.0	0.00		low
resnet50	student_l	pregap	mse_ce	30	29	51.82	51.67	0.15	1.5133	1.5218	3.4	25.00	16.20	medium
resnet50	student_m	postgap	mse	30	29	0.00	0.00	0.00	0.0880	0.0881	0.1	0.00		low
resnet50	student_m	postgap	mse_ce	30	30	13.11	13.11	0.00	0.1632	0.1632	0.7	0.30	-2.58	low
resnet50	student_m	pregap	mse	30	30	0.00	0.00	0.00	1.1344	1.1344	0.0	0.00		low
resnet50	student_m	pregap	mse_ce	30	29	39.24	39.03	0.21	1.5202	1.5197	1.5	15.08	23.55	medium
resnet50	student_s	postgap	mse	30	30	0.00	0.00	0.00	0.0887	0.0887	0.0	0.00		low
resnet50	student_s	postgap	mse_ce	30	28	15.45	15.06	0.39	0.1604	0.1613	7.5	1.92	8.67	low
resnet50	student_s	pregap	mse	30	30	0.00	0.00	0.00	1.2330	1.2330	0.0	0.00		low
resnet50	student_s	pregap	mse_ce	30	30	27.03	27.03	0.00	1.5075	1.5075	0.7	6.29	20.25	low

Critical warnings

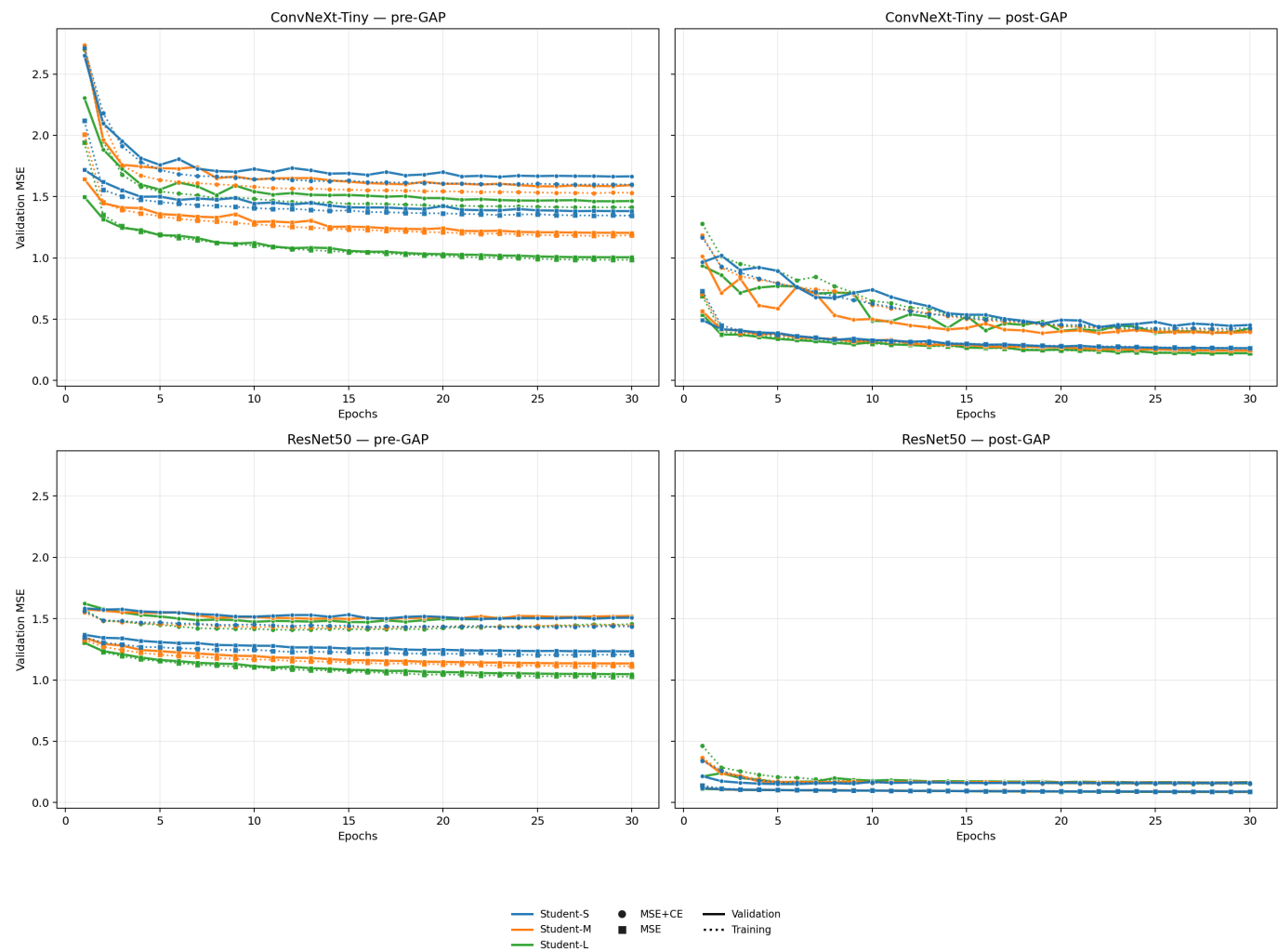
- **convnext_tiny/student_l/pregap/mse_ce (medium)**: train-val Top-1 gap 24.0 pp (overfitting)
- **resnet50/student_l/pregap/mse_ce (medium)**: train-val Top-1 gap 25.0 pp (overfitting)
- **resnet50/student_m/pregap/mse_ce (medium)**: train-val Top-1 gap 15.1 pp (overfitting)

Training curves

Top-1 (4 panels, teacher × target). Grey lines are teacher and CE-baseline references; coloured lines are distilled students. MSE-only runs with placeholder zero Top-1 are omitted here and shown in the MSE plot.



Validation MSE for all distilled runs (lower = better matching).



Distillation — food101

- Best classification run: `convnext_tiny/student_l/pregap/mse_ce` — 62.931% val Top-1.
- Lowest-MSE run: `convnext_tiny/student_l/postgap/mse` — 0.0331 val MSE.

All distilled runs

Teacher	Student	Target	Loss	Ep	Best@	Best T1	Fin T1	T1 drop	Best MSE	Fin MSE	MSE reg%	T1 gap	vs base	Concern
convnext_tiny	student_l	postgap	mse	30	29	0.00	0.00	0.00	0.0331	0.0331	0.0	0.00		low
convnext_tiny	student_l	postgap	mse_ce	30	29	40.58	40.45	0.13	0.0477	0.0476	0.0	-5.51	-24.08	low
convnext_tiny	student_l	pregap	mse	30	30	0.00	0.00	0.00	0.2695	0.2695	0.0	0.00		low
convnext_tiny	student_l	pregap	mse_ce	30	29	62.93	62.46	0.48	0.3422	0.3403	5.4	8.18	-1.73	low
convnext_tiny	student_m	postgap	mse	30	30	0.00	0.00	0.00	0.0353	0.0353	0.0	0.00		low
convnext_tiny	student_m	postgap	mse_ce	30	27	38.01	37.80	0.21	0.0483	0.0479	0.3	-4.56	-15.87	low
convnext_tiny	student_m	pregap	mse	30	29	0.00	0.00	0.00	0.2843	0.2843	0.0	0.00		low
convnext_tiny	student_m	pregap	mse_ce	30	30	58.53	58.53	0.00	0.3392	0.3392	4.4	2.02	4.66	low
convnext_tiny	student_s	postgap	mse	30	26	0.00	0.00	0.00	0.0381	0.0383	0.5	0.00		low
convnext_tiny	student_s	postgap	mse_ce	30	29	35.62	35.60	0.01	0.0478	0.0478	0.1	-5.65	0.04	low
convnext_tiny	student_s	pregap	mse	30	29	0.00	0.00	0.00	0.2954	0.2954	0.0	0.00		low
convnext_tiny	student_s	pregap	mse_ce	30	29	49.39	49.39	0.00	0.3284	0.3283	1.4	-2.58	13.81	low

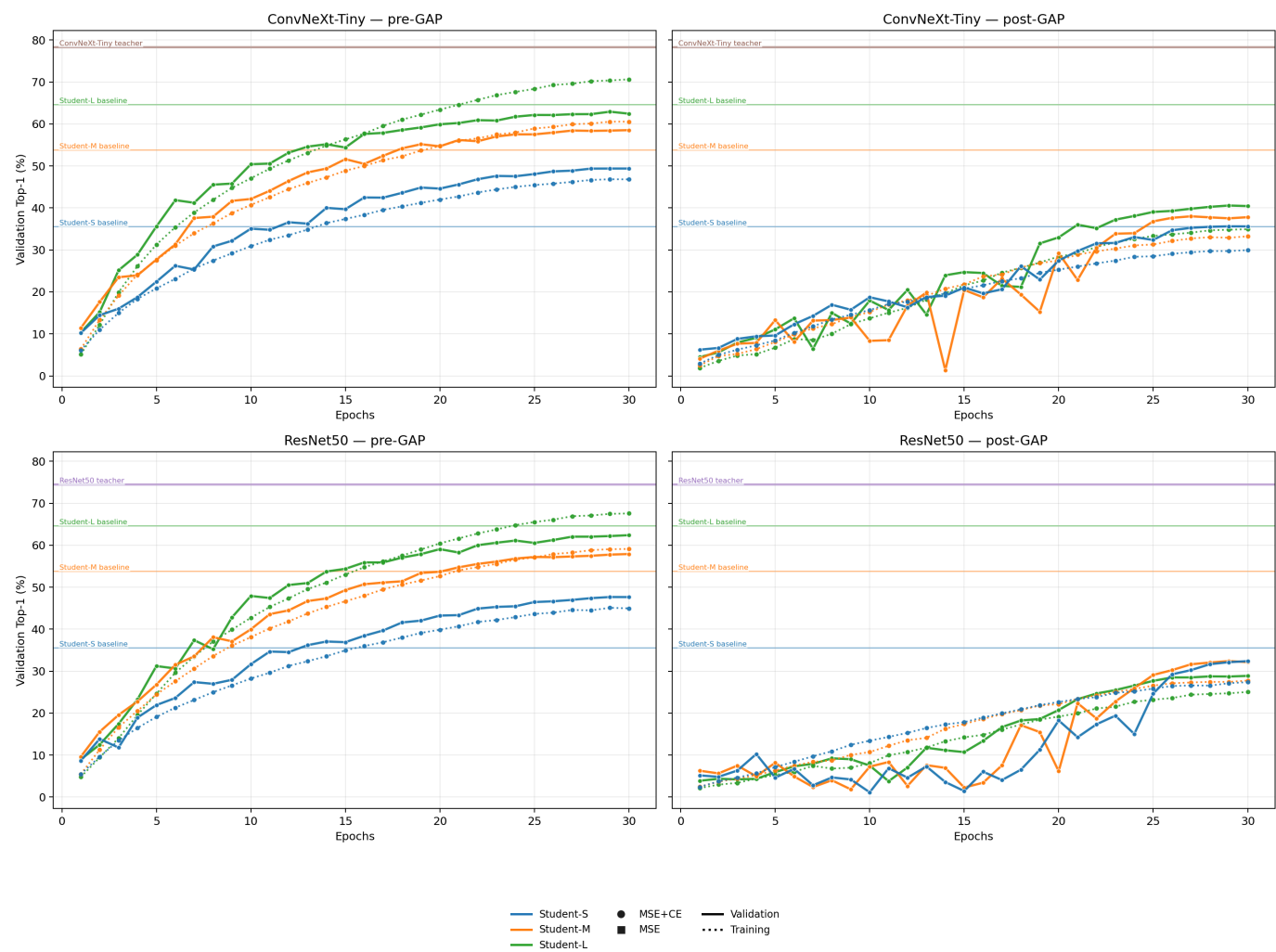
Teacher	Student	Target	Loss	Ep	Best@	Best T1	Fin T1	T1 drop	Best MSE	Fin MSE	MSE reg%	T1 gap	vs base	Concern
resnet50	student_l	postgap	mse	30	30	0.00	0.00	0.00	0.0701	0.0701	0.0	0.00		low
resnet50	student_l	postgap	mse_ce	30	30	28.87	28.87	0.00	0.1163	0.1163	0.1	-3.84	-35.79	low
resnet50	student_l	pregap	mse	30	30	0.00	0.00	0.00	0.8522	0.8522	0.0	0.00		low
resnet50	student_l	pregap	mse_ce	30	30	62.39	62.39	0.00	1.0492	1.0492	0.0	5.18	-2.27	low
resnet50	student_m	postgap	mse	30	29	0.00	0.00	0.00	0.0755	0.0769	1.8	0.00		low
resnet50	student_m	postgap	mse_ce	30	29	32.40	32.15	0.25	0.1165	0.1163	0.2	-4.39	-21.48	low
resnet50	student_m	pregap	mse	30	30	0.00	0.00	0.00	0.9245	0.9245	0.0	0.00		low
resnet50	student_m	pregap	mse_ce	30	30	57.90	57.90	0.00	1.0586	1.0586	0.1	1.20	4.03	low
resnet50	student_s	postgap	mse	30	28	0.00	0.00	0.00	0.0810	0.0814	0.4	0.00		low
resnet50	student_s	postgap	mse_ce	30	30	32.38	32.38	0.00	0.1164	0.1164	0.4	-5.00	-3.19	low
resnet50	student_s	pregap	mse	30	29	0.00	0.00	0.00	0.9735	0.9735	0.0	0.00		low
resnet50	student_s	pregap	mse_ce	30	29	47.66	47.64	0.01	1.0561	1.0565	0.0	-2.69	12.08	low

Critical warnings

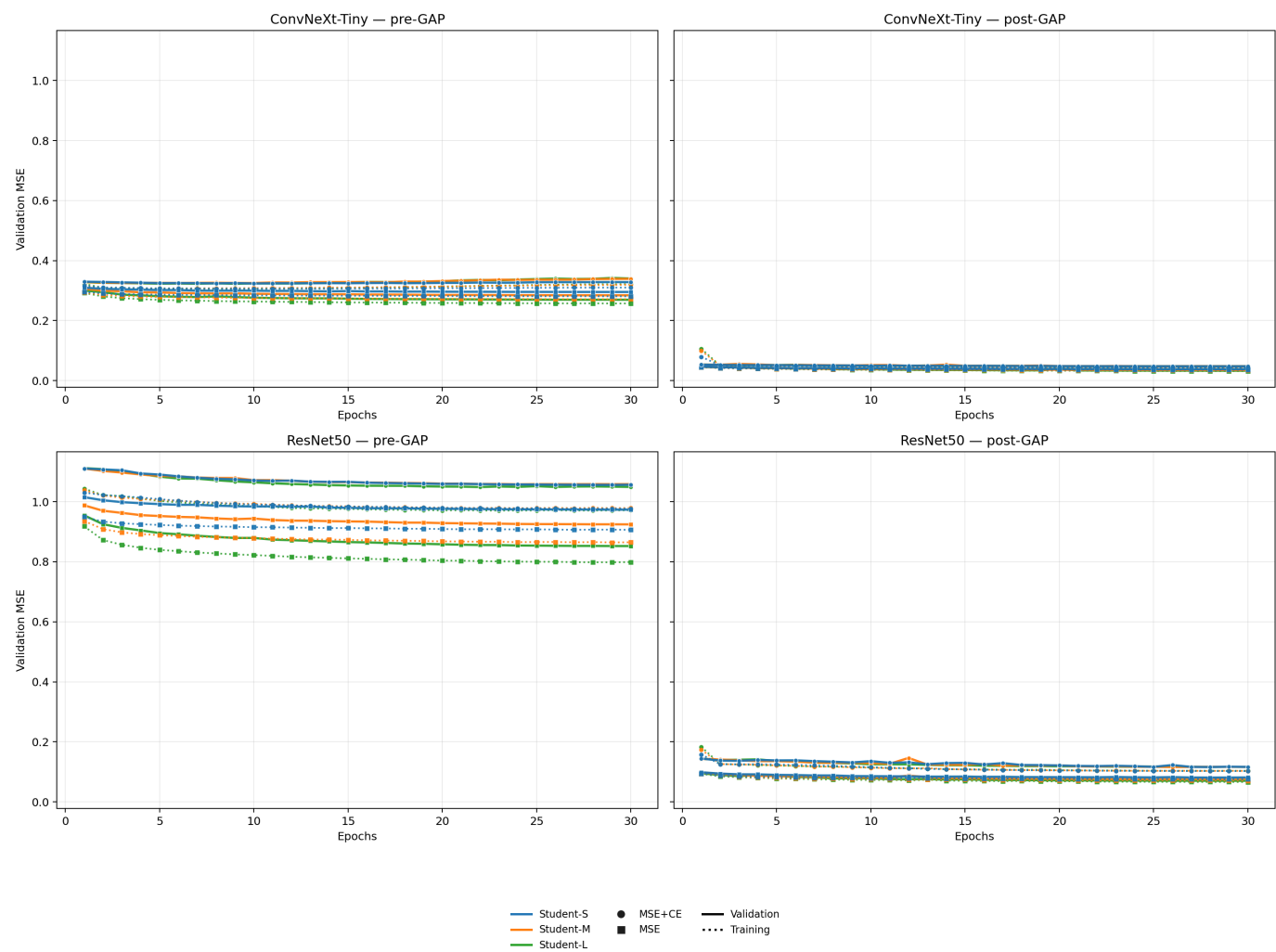
No critical training problems were detected.

Training curves

Top-1 (4 panels, teacher × target). Grey lines are teacher and CE-baseline references; coloured lines are distilled students. MSE-only runs with placeholder zero Top-1 are omitted here and shown in the MSE plot.



Validation MSE for all distilled runs (lower = better matching).



Global Factor Summary

Averaged across both datasets. Top-1 columns use classification-bearing runs (CE / MSE+CE); MSE columns use all distilled runs with MSE logs.

Factor	Value	Mean T1	Best T1	Mean MSE	Best MSE	N
Teacher	ConvNeXt-Tiny	39.18	62.93	0.5151	0.0331	24
Teacher	ResNet50	34.60	62.39	0.6331	0.0701	24
Target	post-GAP	25.76	40.58	0.1470	0.0331	24
Target	pre-GAP	48.02	62.93	1.0013	0.2695	24
Student	Student-L	41.84	62.93	0.5417	0.0331	16
Student	Student-M	38.08	58.53	0.5757	0.0353	16
Student	Student-S	30.75	49.39	0.6050	0.0381	16
Loss	MSE			0.4971	0.0331	24
Loss	MSE+CE	36.89	62.93	0.6512	0.0477	24